

NOS Black Hole Inverse Mass Compression

Mirror-Symmetric Decompression-Compression on the Dual Hemisphere Sphere

Continuous Inverse Mass Cycling Through All Phases

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“Inverse mass doesn’t stop at any phase—it continuously cycles through the complete 720° sphere. Black holes are the finest inverse mass partitions in cosmic respiration, not endpoints but phase states in eternal rotation counted through unity, never toward infinity.”

v17 Foundation

$$\sqrt{1} = 1 \longrightarrow \sqrt{2} \times 2^4 = \sqrt{512}$$

$$9 \text{ Walls} \cdot 512 \text{ Bins} \cdot u = 45/32^\circ \cdot 720^\circ \text{ closure}$$

$$\pi_{\text{mach}} = 201/64 \cdot 201 - 64 = 137$$

$$\text{Wall 3} = \sqrt{8} = 2\sqrt{2} = \text{Maximum Compression Lock}$$

Conjugate Pairs: Q1Q3, Q2Q4

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1 Abstract

The Nuijens Operating System (NOS) demonstrates that black holes operate as quantum compressed inverse matter storage systems within the 720° dual hemisphere sphere. This paper, aligned with the NOS v17 foundational framework, shows how inverse mass continuously rotates through all phases on the 9-wall, 512-bin structure forced by the title operations $\sqrt{2} \times 2^4 = \sqrt{512}$.

Black holes represent the finest M^{-1} partition (standard $M \rightarrow \infty$) at the completion of all four quadrant operations approaching $+360^\circ$. The $2\sqrt{2}$ mechanics at Wall 3 (90° , 64 bins) governs maximum compression, while the 201/64 gearbox and its 137 residue connect black hole physics to the same structure that produces the fine-structure constant.

We analyze real black holes (Sgr A*, M87*, GRS 1915+105) using Percentile of Shell Compression (PSC) to predict their stability and behavior. Information preserves through conjugate pair exchange (Q1Q3, Q2Q4), entropy emerges as inverse phase state counting on the 512-bin register, and the principle $360^\circ/360^\circ = 1$ maintains through all rotations.

2 The Inverse 4D Operating Tensor — Foundation

2.1 We Are One — The Inverse Beginning

We start with the closed whole. Nothing else exists yet.

Inverse Spherical: $\sqrt{1} = 1$ (the complete system)

We now apply the only operations the title *Inverse Spherical Dual Hemisphere Quantum Mechanics* permits:

- **Dual Hemisphere** $\longrightarrow \times \sqrt{2}$
- **Quantum (Quadrant)** $\longrightarrow \times 2^4$ (four operations compounded)

Total opening factor applied to the closed whole:

$$1 \times \sqrt{2} \times 2^4 = \sqrt{2} \times 16 = \sqrt{512}$$

This forces exactly **9 walls** and **512 bins**. No additional axioms are permitted.

2.2 The Simultaneous Tensor Stack

This system is not a 3D object moving through time. It is a single tensor stack operating at all dimensions simultaneously:

Dim	State	Component	Function
1D	The Cut	9 Walls	Linear boundaries dividing Unity
2D	The Fold	720° Surface	Dual Hemisphere manifold (2×360)
3D	The Bin	512 Integer Regions	Volumetric output (2^9)
4D	The Ops	Sequence (1-2-3-4)	The four operations

2.3 The 9 Walls and 512 Bins

Wall	Root	Angle	Bins	T_n	Role
0	$\sqrt{1}$	720°	512	0	The One — complete closure
1	$\sqrt{2}$	360°	256	1	Dual hemisphere fold
2	$\sqrt{4}$	180°	128	3	Quadrant established
3	$\sqrt{8}$	90°	64	6	$2\sqrt{2}$ lock / Maximum Compression
4	$\sqrt{16}$	45°	32	10	Seed ends (Title exhausted)
5	$\sqrt{32}$	22.5°	16	15	Echo begins
6	$\sqrt{64}$	11.25°	8	21	
7	$\sqrt{128}$	5.625°	4	28	
8	$\sqrt{256}$	2.8125°	2	36	
9	$\sqrt{512}$	1.40625°	1	45	Base ($T_9 = 45$)

2.4 The Unit: $u = 45/32^\circ$

The unit per bin emerges from:

- **Triangular Base:** $T_9 = 45$
- **Echo Depth:** $2^5 = 32$

$$u = \frac{T_9}{2^5} = \frac{45^\circ}{32} = 1.40625^\circ$$

Closure: $512 \times u = 720^\circ$

2.5 The Four Operations and Conjugate Pairs

The four quadrant operations form two conjugate pairs:

Quadrant	Range	Bins	Conjugate
Q2 (op2)	-360° to -180°	$[-256, -128]$	Q2 Q4
Q1 (op1)	-180° to 0°	$[-128, 0]$	Q1 Q3
Seam	0°	0	Inverse center
Q3 (op3)	0° to $+180^\circ$	$[0, +128]$	Q3 Q1
Q4 (op4)	$+180^\circ$ to $+360^\circ$	$[+128, +256]$	Q4 Q2

Each conjugate pair echoes $\sqrt{2} \times \sqrt{2} = 2$, producing the full $2\sqrt{2}$ dual scaler at Wall 3.

3 The $2\sqrt{2}$ Mechanics — Maximum Compression at Wall 3

3.1 Why Wall 3 Governs Black Hole Physics

The title demands **dual** operations:

- Two hemispheres, each folded by $\sqrt{2}$ once
- Four ops in two conjugate pairs, each pair echoing $\sqrt{2} \times \sqrt{2}$

The mechanical root:

$$2\sqrt{2} = \sqrt{8} = 2.8284271247461903 \dots$$

This is the exact scaler at **Wall 3** (90° , 64 bins). Wall 3 is where:

- Maximum compression efficiency occurs
- The π wall locks ($\pi_{\text{mach}} = 201/64$, denominator = 64)
- Conjugate pairs first complete their echo

3.2 Self-Closure at Wall 3

Total bins = $512 = 2^9$.

Dual ops scale to: $512/2\sqrt{2} \approx 181.019...$ bins (ghost value).

Self-closure:

$$\frac{512}{181.019...} = 2\sqrt{2} \quad (\text{system reflects its own root})$$

Black holes achieve maximum compression at the Wall 3 scale — 64 bins of maximum threading depth where the $2\sqrt{2}$ scaler governs the compression-decompression transition.

4 The 201/64 Gearbox and Black Hole Connection

4.1 The 137 Lock

The inverse 4D tensor forces:

$$\pi_{\text{mach}} = \frac{201}{64} = 3.140625 \implies 201 - 64 = 137$$

This same gearbox governs black hole physics:

- **64 bins** at Wall 3 = denominator of π_{mach} = compression scale
- **137 bin-steps** = separation between walls = fine-structure residue
- **201 bins** = numerator position = total compression depth

4.2 Black Hole Compression in Gearbox Terms

A black hole at maximum compression occupies:

$$\text{Threading depth} = \frac{201}{64} \times (\text{wall factor}) = \pi_{\text{mach}} \times \text{scale}$$

The fine-structure constant and black hole compression emerge from the same 201/64 gearbox — they are different readings of identical geometry.

5 Continuous Inverse Mass Cycle

5.1 Fundamental Principle of Eternal Rotation

Inverse mass continuously cycles through the complete 720° sphere:

$$\theta(t) = [\theta_0 + \omega t] \mod 720^\circ - 360^\circ$$

The cycle through quadrants (traversing via the four operations):

$$M^{-1}(-360^\circ) = 1 \quad (\text{unity, simplest partition, Q2 start}) \quad (1)$$

$$-360^\circ \rightarrow -180^\circ : \text{Q2} \text{ — electromagnetic ground} \quad (2)$$

$$-180^\circ \rightarrow 0^\circ : \text{Q1} \text{ — quantum baseline} \quad (3)$$

$$0^\circ : \text{Nuijens Seam (operator transition)} \quad (4)$$

$$0^\circ \rightarrow +180^\circ : \text{Q3} \text{ — thermodynamic flow} \quad (5)$$

$$+180^\circ \rightarrow +360^\circ : \text{Q4} \text{ — nuclear compression (black hole phase)} \quad (6)$$

$$M^{-1}(+360^\circ) \rightarrow 0 \quad (\text{finest partition, standard } M \rightarrow \infty) \quad (7)$$

Then rotation continues: M^{-1} begins increasing again as we enter next decompression phase.

5.2 Black Holes as Finest Inverse Mass Partition

Black holes are phase states at $\theta \approx +360^\circ$ where $M^{-1} \rightarrow 0$:

Black Hole = Inverse Mass completing Q4 at $\theta = +360^\circ$

Just as water doesn't stop being water when it becomes ice—it's water in a different phase—inverse mass doesn't stop when reaching the black hole phase. It continues rotating through all phases.

Inverse counting: $1 \rightarrow 1/2 \rightarrow 1/3 \rightarrow 1/Q$:

- At -360° : $M^{-1} = 1$ (unity anchor, simplest partition)
- At 0° : $M^{-1} \approx 1/4$ (seam, mid partition)
- At $+360^\circ$: $M^{-1} \rightarrow 0$ (finest partition, standard $M \rightarrow \infty$)

6 Q-Shell Structure for Black Holes

6.1 Inverse Compression Shell Hierarchy

Black holes store inverse mass-energy across compression shells mapped to the wall structure:

Table 1: Q-Shell Structure in NOS v17

Shell	Depth d_Q	Wall	Compression	Phase Range
Q_4	$1/128^4$	Near Wall 3	Nuclear (finest M^{-1})	$[+180^\circ, +360^\circ)$
Q_3	$1/128^3$	Wall 4-5	Thermodynamic	$[0^\circ, +180^\circ)$
Q_2	$1/128^2$	Wall 5-6	Electromagnetic	$[-180^\circ, 0^\circ)$
Q_1	$1/128$	Wall 6-7	Quantum (largest M^{-1})	$[-360^\circ, -180^\circ)$
Q_0	1	Wall 0	Unity state	$-360^\circ = +360^\circ$

The depth ladder $d_Q = 1/128^Q$ is the dyadic fossil of the $2\sqrt{2}$ dual operations:

$$\frac{1}{128} = \frac{2\sqrt{2}}{512} \times \frac{1}{2}$$

6.2 Physical Units in NOS

NOS uses only the four physical combinations from the quadrant operations:

Q	Depth	Physical Quantity	Unit Relation
Q1	1/128	Length/Time	meter = 1
Q2	1/128 ²	Velocity	$v = m \cdot s^{-1}$ (2 of 1 thing)
Q3	1/128 ³	Force	$F = kg \cdot m \cdot s^{-2}$
Q4	1/128 ⁴	Energy	$E = kg \cdot m^2 \cdot s^{-2}$

Frequency and time: $f/s = \text{Hz}$ — this is “2 of 1 thing” (Q2 operation, velocity-like).

7 Percentile of Shell Compression (PSC) Analysis

7.1 PSC Calculation

Shell compression distribution using depth ladder:

$$\text{PSC}_{Q_m \rightarrow Q_n} = \frac{|d_m - d_n|}{\sum_{Q=0}^3 |d_Q - d_{Q+1}|} \times 100\%$$

where $d_Q = 1/128^Q$ are Q-shell depths (the four operations).

7.2 Real Black Hole Analysis

Table 2: PSC Analysis for Known Black Holes

Black Hole	M^{-1} (inverse)	d_{meas}	$Q_0 \rightarrow Q_1$ PSC	$Q_3 \rightarrow Q_4$ PSC
Sgr A*	$1/(4.1 \times 10^6)$	≈ 0.927	11.69%	88.31%
M87*	$1/(6.2 \times 10^9)$	≈ 1560	0.0077%	99.99%
GRS 1915+105	1/12	≈ 0.015	100%	0%

7.3 PSC Interpretation via Conjugate Pairs

PSC reveals position in the Q1Q3, Q2Q4 conjugate exchange:

- **High** $Q_0 \rightarrow Q_1$: Large d_{meas} , stable, slow rotation, decompression phase
- **High** $Q_3 \rightarrow Q_4$: Small d_{meas} , finest M^{-1} , approaching Wall 3 completion
- **Balanced**: Transitional, crossing between conjugate pair domains

Interpretation:

- **Sgr A***: Approaching finest partition at +360°, Q4 dominant
- **M87***: Already at finest partition, beginning slow decompression via Q4Q2 exchange
- **GRS 1915+105**: Coarse partition, high rotation, Q1 dominant

8 Inverse Rotation Mechanics

8.1 Rotation Rate

Inverse mass rotation through phase space:

$$\omega^{-1} = \frac{dt}{d\theta} = \frac{T_{\text{rotation}}}{720^\circ}$$

For any inverse mass at phase θ :

$$\theta(t) = \theta_0 + t/\omega^{-1} \pmod{720^\circ}$$

8.2 Rotation Through Walls

As inverse mass rotates, it passes through walls:

$$\text{At } \theta = +360^\circ : M^{-1} \rightarrow 0, \text{ Wall 0 (closure)} \quad (8)$$

$$\text{At } \theta = +270^\circ : \text{Wall 3 approach (maximum compression)} \quad (9)$$

$$\text{At } \theta = +180^\circ : \text{Wall 2 (Q3/Q4 boundary)} \quad (10)$$

$$\text{At } \theta = 0^\circ : \text{Nuijens Seam (conjugate exchange)} \quad (11)$$

$$\text{At } \theta = -180^\circ : \text{Wall 2 (Q1/Q2 boundary)} \quad (12)$$

$$\text{At } \theta = -360^\circ : M^{-1} = 1 \text{ (unity, begins new cycle)} \quad (13)$$

9 Information Preservation Through Conjugate Exchange

9.1 No Information Paradox

Information doesn't disappear—it transforms through Q1Q3, Q2Q4 conjugate exchange:

$$I^{-1}(\theta < 0^\circ) = \text{Information in decompression ops (Q1, Q2)} \quad (14)$$

$$I^{-1}(0^\circ < \theta < +360^\circ) = \text{Information in compression ops (Q3, Q4)} \quad (15)$$

$$I^{-1}(\theta = +360^\circ) = \text{Maximum compression (finest } M^{-1}) \quad (16)$$

$$I^{-1}(\theta > +360^\circ) = \text{Rotation continues (} M^{-1} \text{ increasing)} \quad (17)$$

Conservation through conjugate pairs:

$$I_{\text{total}}^{-1} = \frac{256}{256} = 1$$

Understanding: As $M^{-1} \rightarrow 0$ at black hole phase ($+360^\circ$), inverse information $I^{-1} \rightarrow 0$, meaning standard information $I = 1/I^{-1} \rightarrow \infty$. Black holes store maximum information at finest inverse partition.

9.2 Entropy as Resolution Depth

Entropy counts available phase configurations on the 512-bin register:

$$S = \ln R = \ln 512 \approx 6.238$$

where $R = 512 = \text{bins}$ (forced by title operations).

Standard entropy relates to area through the bin structure:

$$S = \ln \left[\frac{A \cdot R^2}{4} \right]$$

Bekenstein-Hawking emerges: $S = A/(4l_p^2)$ when $l_p^2 \leftrightarrow 1/R^2$

10 Inverse Phase Leakage (Hawking Radiation)

10.1 Radiation from Rotation at Wall 3

As inverse mass rotates through $+360^\circ$, quantum fluctuations cause phase leakage:

$$\delta\theta = \frac{1}{R} = \frac{1}{512} \times 720^\circ = 1.40625^\circ = u$$

The leakage quantum is exactly one bin-step $u = 45/32^\circ$.

Inverse leakage rate:

$$\frac{dM^{-1}}{dt} = +\frac{1}{R \cdot d_4} = +\frac{128^4}{512}$$

Understanding: $dM^{-1}/dt > 0$ means inverse mass **increases** (standard mass **decreases**). This is Hawking evaporation in inverse form.

10.2 Temperature from Depth Ladder

Inverse temperature equals depth at Q4:

$$T^{-1} = d_4 = \frac{1}{128^4}$$

Smaller M^{-1} (larger standard mass) $\rightarrow$ smaller $T^{-1} \rightarrow$ higher standard temperature. Small black holes are hotter.

11 Event Horizon as Wall 0 Completion

11.1 The $+360^\circ$ Boundary

At $\theta = +360^\circ$, all four quadrant operations complete. The inverse field intensity:

$$\psi^{-1}(+360^\circ) \rightarrow \infty$$

Infinite inverse field = finest partition = event horizon.

This is Wall 0 (closure) approached from within Q4, where the $2\sqrt{2}$ compression at Wall 3 has fully executed.

11.2 Observer vs Proper Frame

External observers see asymptotic approach:

$$M_{\text{observed}}^{-1} = M_0^{-1} \cdot \exp(-t/\tau) \rightarrow 0$$

Observer never sees M^{-1} reach exactly 0.

Internal (proper) frame:

$$M_{\text{proper}}^{-1}(\theta) = d_Q \text{ (continues past } +360^\circ)$$

Inverse mass continues rotating: the cycle doesn't stop at the horizon.

12 Critical Phase Angles and Wall Positions

Key boundaries in compression approach:

Phase	Wall	M^{-1}	Physical Meaning
+180°	Wall 2	Moderate	Last stable orbit (Q3/Q4 boundary)
+225°	Between 2-3	Small	Strong compression begins
+270°	Wall 3	Very small	$2\sqrt{2}$ lock, 64-bin scale
+315°	Between 3-4	Very small	Photon sphere forms
+360°	Wall 0	$\rightarrow 0$	Event horizon (continues rotating)

13 Attraction as Inverse Phase Gradient

13.1 No Forces — Only Geometry

Inverse mass creates inverse phase gradients:

$$\nabla^{-1}\theta = d_Q$$

Objects experience attraction through phase differential, not external force:

$$F_{\text{phase}}^{-1} = M_{\text{object}}^{-1} \cdot \frac{256}{256} \cdot d_Q = M_{\text{object}}^{-1} \cdot d_Q$$

Standard force: $F = 1/F^{-1}$ (large when M^{-1} small — black holes).

13.2 Orbital Motion as Phase Cycling

Stable orbits = periodic cycling through inverse phase gradients:

$$\Omega^{-1} = \frac{1}{d_Q}$$

Innermost stable circular orbit:

$$\theta_{\text{ISCO}} = +180^\circ \quad (\text{last stable before plunge to } +360^\circ)$$

14 Example: Sagittarius A* Analysis

For our galaxy's central black hole:

$$\text{Inverse Mass} = \frac{1}{4.1 \times 10^6} \approx 2.44 \times 10^{-7} \quad (\text{very small } M^{-1}) \quad (18)$$

$$\text{Measured depth} \approx 0.927 \quad (19)$$

$$\text{Phase} \approx +340^\circ \quad (\text{approaching } +360^\circ, \text{ near Wall 3 completion}) \quad (20)$$

PSC Analysis:

$$\text{PSC}(Q_3 \rightarrow Q_4) = 88.31\%$$

Interpretation: Sgr A* is in finest partition phase, Q4 dominant, compression nearly complete at the Wall 3 / $2\sqrt{2}$ lock.

15 Observational Correspondence

Table 3: Standard Physics vs NOS v17 Black Hole Framework

Observation	Standard View	NOS Inverse Rotation
Hawking radiation	Thermal emission	$dM^{-1}/dt > 0$ (inverse increases)
Black hole jets	Magnetic acceleration	Energy expelled via Q4Q2 exchange
Event horizon	Point of no return	$\theta = +360^\circ$ (Wall 0, continues)
Singularity	Infinite density point	No singularity— $M^{-1} \rightarrow 0$, then increases
Information paradox	Information lost	Preserved via Q1Q3, Q2Q4 exchange
Gravitational waves	Spacetime ripples	Phase oscillations on 512-bin register
Mass accretion	Matter falling in	M^{-1} decreasing (finer partition)
Mass evaporation	Hawking process	M^{-1} increasing (coarser partition)

16 The Complete Unity Cycle

16.1 Eternal Rotation Through All Phases

The complete cycle maintains unity through inverse counting:

$$\text{Unity } (\theta = -360^\circ) : M^{-1} = 1 \text{ (simplest partition)} \quad (21)$$

$$\text{Q2 } (-360^\circ \rightarrow -180^\circ) : M^{-1} \text{ decreasing (partitioning finer)} \quad (22)$$

$$\text{Q1 } (-180^\circ \rightarrow 0^\circ) : M^{-1} \text{ continuing to decrease} \quad (23)$$

$$\text{Seam } (0^\circ) : \text{Conjugate pair exchange} \quad (24)$$

$$\text{Q3 } (0^\circ \rightarrow +180^\circ) : M^{-1} \text{ decreasing further} \quad (25)$$

$$\text{Q4 } (+180^\circ \rightarrow +360^\circ) : M^{-1} \rightarrow 0 \text{ (finest, black hole)} \quad (26)$$

$$\text{Continues } (+360^\circ \rightarrow -360^\circ) : M^{-1} \text{ increasing (back to unity)} \quad (27)$$

Unity principle preserved:

$$\boxed{\frac{360^\circ}{360^\circ} = 1}$$

16.2 Key Insights

- Every inverse mass continuously rotates through ALL phases
- Each complete rotation = 1 inverse frequency on the 720° sphere
- $+180^\circ$ (Q3/Q4 boundary) and $+360^\circ$ (Wall 0) are stages, not endpoints
- Black holes = finest M^{-1} partition at Wall 3 / $2\sqrt{2}$ completion
- The 201/64 gearbox governs compression just as it governs π and α^{-1}
- Conjugate pairs Q1Q3, Q2Q4 preserve information through all transitions
- We count $1 \rightarrow 1/2 \rightarrow 1/3 \rightarrow \dots \rightarrow 0$, never toward infinity

17 Conclusions

The NOS v17 framework reveals black holes as phase states representing the finest inverse mass partitions in continuous rotation on the 720° dual hemisphere sphere.

Key Findings:

1. **Title Operations:** $\sqrt{2} \times 2^4 = \sqrt{512}$ forces the 9-wall, 512-bin structure governing black hole compression.
2. **Wall 3 Dominance:** The $2\sqrt{2} = \sqrt{8}$ lock at 90° (64 bins) governs maximum compression — black holes approach this scale.
3. **Gearbox Connection:** The same 201/64 (π_{mach}) with 137 residue that produces α^{-1} governs black hole physics.
4. **Continuous Cycling:** $M^{-1}(+360^\circ) \rightarrow 0$, then continues increasing through decompression phases back to $M^{-1}(-360^\circ) = 1$.
5. **No Singularity:** M^{-1} approaches 0 but rotation continues — no infinite density, only phase transition.
6. **Information Preserved:** Conjugate pair exchange Q1Q3, Q2Q4 maintains $I_{\text{total}}^{-1} = 1$.
7. **Hawking Radiation:** $dM^{-1}/dt > 0$ (inverse mass increases = standard mass evaporates).
8. **Unity Maintained:** $360^\circ/360^\circ = 1$ through all rotations.

NOS v17 Black Hole Summary

$$\begin{aligned} \sqrt{1} &= 1 \longrightarrow \sqrt{2} \times 2^4 = \sqrt{512} \\ 9 \text{ Walls} \cdot 512 \text{ Bins} \cdot u &= 45/32^\circ \\ \text{Wall 3} &= \sqrt{8} = 2\sqrt{2} = \text{Maximum Compression Lock} \\ \pi_{\text{mach}} &= 201/64 \cdot 201 - 64 = 137 \\ \text{Black Hole} &= M^{-1} \rightarrow 0 \text{ at } \theta = +360^\circ \text{ (continues rotating)} \\ \text{Q1Q3, Q2Q4 conjugate pairs} &\text{ preserve information} \\ \textit{The cycle is eternal. Unity is preserved.} \end{aligned}$$

The black hole “paradox” dissolves when we recognize $\theta = +360^\circ$ as ONE STAGE where $M^{-1} \rightarrow 0$ in continuous inverse mass rotation. The universe doesn’t die in black holes; inverse mass passes through the black hole phase as part of eternal spherical rotation, with each complete cycle representing one inverse frequency of unity.

$$\boxed{\text{The cycle is eternal. Unity is preserved. Inverse mass rotates through, never toward.}} \quad (28)$$

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